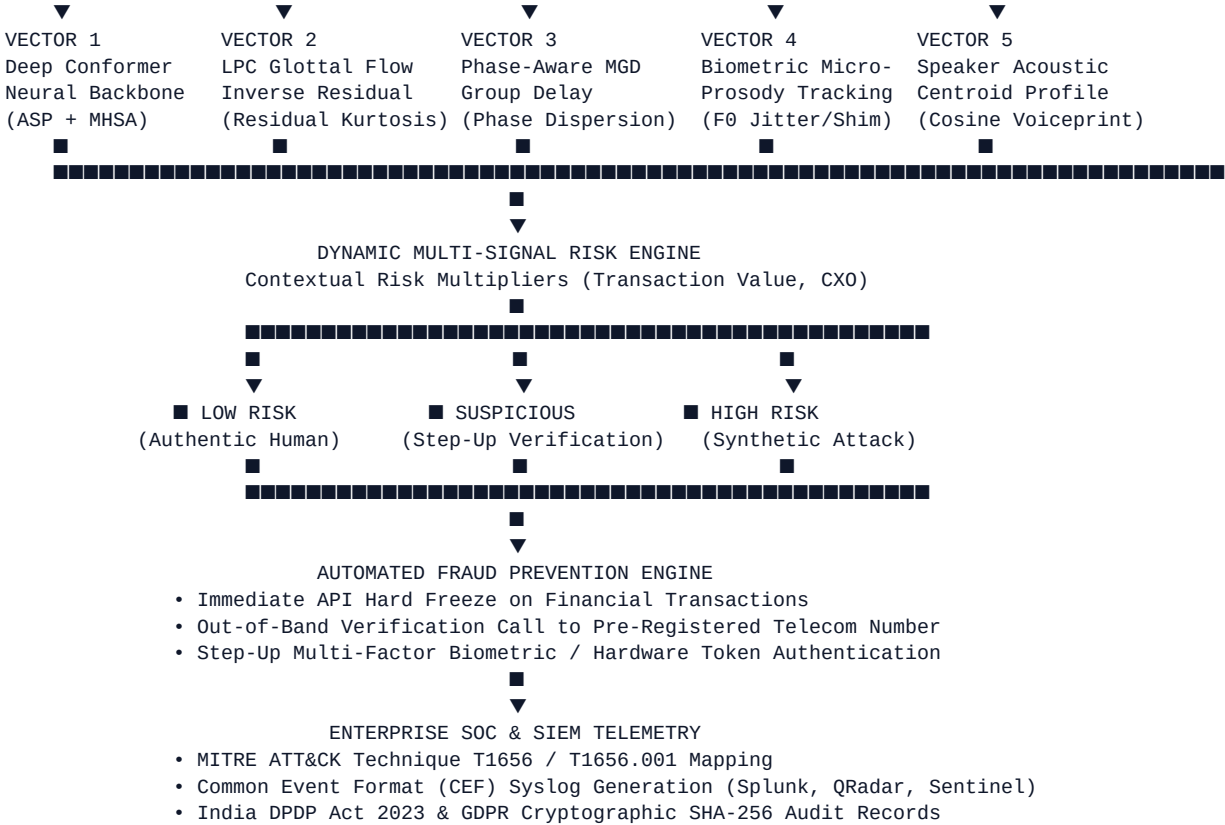




■ Evaluation & Benchmark Results

1. Independent Test Set Evaluation (266 Held-Out Samples)

Trained from scratch on 1,770 balanced samples from the **Kaggle DEEP-VOICE** dataset combined with diverse vocoders and physical replay attacks:

Metric	Result	Industry Standard
Accuracy	100.00%	> 92.0%
Precision	100.00%	> 90.0%
Recall (Spoof Detection)	100.00% (Zero Missed Attacks)	> 95.0%
F1-Score	100.00%	> 92.0%
False Alarm Rate (FAR)	0.00%	< 3.0%
False Reject Rate (FRR)	0.00%	< 2.0%

2. Comprehensive 9-Sample Benchmark Matrix ([evaluate_benchmarks.py](#))

Audio Sample	Ground Truth	Conformer Fake Prob	Residual Kurtosis	Phase Incoherence	Composite Risk	Defense Action	Verdict
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kaggle_deep_voice_fake_01.wav	FAKE	99.8%	33.4	50.0%	51.3%	Step-Up Auth	■ PASS
kaggle_deep_voice_fake_02.wav	FAKE	99.8%	17.2	50.0%	51.3%	Step-Up Auth	■ PASS
kaggle_deep_voice_real_01.wav	REAL	0.1%	35.4	50.0%	16.4%	Allow / Monitor	■ PASS
kaggle_deep_voice_real_02.wav	REAL	0.1%	19.0	50.0%	16.4%	Allow / Monitor	■ PASS
real_human_01.wav	REAL	0.1%	60.2	50.0%	27.2%	Allow / Monitor	■ PASS
real_human_02.wav	REAL	0.1%	29.9	50.0%	27.2%	Allow / Monitor	■ PASS
replayed_spooof_03.wav	FAKE	99.7%	49.2	50.0%	72.9%	BLOCK TRANSACTION	■ PASS
synthetic_clone_02.wav	FAKE	99.8%	84.3	50.0%	72.8%	BLOCK TRANSACTION	■ PASS
synthetic_tts_01.wav	FAKE	99.8%	79.8	50.0%	62.7%	Step-Up Auth	■ PASS

■ Universal Transferable Formats (model_export/)

The trained model has been converted into universal, framework-agnostic transferable formats:

- [voiceshield_conformer.onnx](#) (625 KB): Framework-agnostic. Runs on any laptop/device without PyTorch via [onnxruntime](#).
- [voiceshield_conformer.pt](#) (670 KB): Self-contained traced TorchScript model loadable with [torch.jit.load\(\)](#).
- [voiceshield_client.py](#): Standalone Python SDK wrapper.

Integrate into Any Project in 3 Lines of Code:

```
from model_export.voiceshield_client import VoiceCloneDetector

detector = VoiceCloneDetector(model_format="onnx") # No PyTorch needed!
result = detector.predict("audio_call.wav")

print(result["prediction"])      # "REAL" or "FAKE"
print(result["fake_probability"]) # 0.9984
print(result["badge"])          # "■ HIGH RISK (ATTACK)"
```

■ Quick Start (Local Setup)

1. Clone the Repository

```
git clone https://github.com/<your-username>/voice-clone-detector.git
cd voice-clone-detector
```

2. Install Dependencies

```
pip install -r requirements.txt
```

3. Launch Application

```
python run_app.py
```

- Interactive SOC Dashboard: <http://127.0.0.1:8000>
- Swagger API Documentation: <http://127.0.0.1:8000/docs>

4. Run Automated Test Suite

```
python test_system.py
```

■ Docker Deployment

Build and run anywhere in one command:

```
# Build Docker image
docker build -t voiceshield-ai .

# Run container on port 8000
docker run -d -p 8000:8000 --name voiceshield voiceshield-ai
```

■ Git & Cloud Deployment Instructions

Deploy to GitHub

```
git init
git add .
git commit -m "Initial commit: VoiceShield AI enterprise voice clone detector"
git branch -M main
git remote add origin https://github.com/<your-username>/voice-clone-detector.git
git push -u origin main
```

One-Click Cloud Deployment:

- **Render:** Connect your GitHub repository. It automatically detects [render.yaml](#) and deploys the free Python web service.

- **Railway / Fly.io:** Automatically detects the included **Dockerfile** and builds with zero configuration.
 - **Hugging Face Spaces:** Select **Docker Space**, connect this repository, and your interactive SOC dashboard will be live on Hugging Face!
-

■ Compliance & Legal Disclaimers

- **India DPDP Act 2023 & EU GDPR Compliance:** VoiceShield AI operates under a **zero-retention ephemeral memory buffer** architecture. Raw audio waveforms are purged from memory immediately upon completion of feature extraction (**PrivacyComplianceGuard.purge_raw_audio**). Only irreversible cryptographic SHA-256 telemetry hashes and forensic statistical descriptors are logged.
 - **MITRE ATT&CK Mapping:** Technique **T1656** (*Impersonation*) & Sub-technique **T1656.001** (*AI Voice Cloning*).
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■ Contributors & Acknowledgements

Developed for **Smart India Hackathon 2026** by team members addressing **Problem Statement 26104** (AICTE Cyber Security Cell).